

Rajesh Kumar Patro

AI/ML Engineer | NLP & Applied Machine Learning

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ABOUT ME

Pre-final year **B.Tech CSE** student (CGPA 8.23/10) focused on AI, Machine Learning, and Computer Vision. Experienced in building deep learning models, image processing pipelines, and practical AI applications using Python, and OpenCV. Eager to contribute to research-oriented environments and industry teams solving real-world engineering challenges through hands-on ML and vision projects.

TECHNICAL SKILLS

- **Programming Languages:** Python, JavaScript, Java, C
- **Data Analysis & Visualization:** NumPy, Pandas, Matplotlib, Seaborn, Exploratory Data Analysis (EDA)
- **Machine Learning:** Supervised & Unsupervised Learning, Regression, Classification, Clustering, Ensemble Methods (Random Forest), Scikit-learn, Model Evaluation Metrics
- **Deep Learning & NLP:** Artificial Neural Networks (ANN), CNN, RNN, LSTM, TensorFlow, Keras, Text Classification
- **Generative AI & LLMs:** Prompt Engineering, LangChain, Retrieval-Augmented Generation (RAG)
- **MLOps & Deployment:** Docker, Git, GitHub Actions, CI/CD Pipelines, Flask, FastAPI
- **Tools & Platforms:** Jupyter Notebook, Google Colab, Git, GitHub, VS Code, Postman

PROFESSIONAL EXPERIENCE

Neuromatch , Research Intern

07/2026 – 07/2026 | Remote

- Completed Neuromatch Academy's 3-week Deep Learning Track, collaborating with an international research team to train vision models on noisy **CIFAR** datasets for the project "**Vision with Lost Glasses**" under dedicated Teaching Assistant (TA) mentorship.

CVIT, IIIT Hyderabad, AI & Computer Vision Research Trainee

07/2026 – 07/2026 | Hyderabad, India

- Completed an intensive AI summer school covering deep learning, Transformers, LLMs, and Vision-Language Models (VLMs). Gained hands-on experience in full-cycle system development, pre-training, and post-training alignment through daily PyTorch coding labs led by **IIIT Hyderabad**, **IIT faculty**, and industry researchers from **Google DeepMind** and **Microsoft Research**.

Indian Institute of Computing and Technology (IICT), AI & ML Intern

06/2026 – 07/2026 | Remote

- Completed an AI & ML internship at IICT, implementing end-to-end supervised and unsupervised learning pipelines, exploratory data analysis, and predictive modeling under the guidance of **Prof. Ashok Gopalakrishnan** and **Ashish Negi** (Scientific Officer).

PROJECTS

Spam Mail Detector

- Trained a Naive Bayes text classification model on a labelled email dataset, achieving > 95% accuracy on the held-out test set.
- Built full ML pipeline: data cleaning → TF-IDF vectorisation → model training → evaluation with precision/recall/F1 metrics. Github link: <https://github.com/rajesh23092023/Spam-mail-detector> 

Whatsapp Chat Analyser

- Parsed exported WhatsApp chats to surface insights - message frequency, word clouds, emoji usage, and timeline trends, deployed live on Streamlit Cloud.
- Engineered a custom Hinglish stopword list for accurate NLP on Indian-language chats, with a modular codebase (preprocessor, helper, UI layers). Github link: <https://github.com/rajesh23092023/Whatsapp-Chat-Analysis> 

Fake News Detection

- Built an end-to-end NLP classification pipeline from scratch on the ISOT dataset, comparing BoW and TF-IDF feature extraction across KNN, Logistic Regression, Random Forest, and MLP classifiers.
- Achieved ~99.8% accuracy with Random Forest, outperforming baseline models; documented methodology and results in a full IEEE two-column report.

AI Chatbot

- Built a full-stack conversational AI interface using Next.js with server-side API routes integrating Google AI Studio (Gemini) as the LLM backend.
- Deployed on Vercel; live at: <https://ai-sand-delta.vercel.app/> 

HACKATHON & CERTIFICATE

BPUT Hackathon 2026 Participant: Developed a real-time computer vision system with a 4-member team to automate workplace safety monitoring; detected PPE compliance (helmets, vests, gear) and hazards (smoke/fire) from video feeds with instant incident alerts.

Python for Data Science, AI & Development: IBM Certification Course

Developing AI Applications with Python and Flask: IBM Certification Course

Developing Back-End Apps with Node.js and Express: IBM Certification Course

EDUCATION

B.Tech in Computer Science and Engineering
Gandhi Academy of Technology and Engineering (GATE) 

06/2024 – Present | Berhampur, Odisha, India

Higher Secondary (+2), Science Stream, *Government Science College,* 

06/2022 – 06/2024 | Chatrapur, Odisha, India

LANGUAGES

- English
- Odia
- Hindi

INTERESTS

- Movies & Web series
- Coding
- Traveling & Exploration